

Manager-worker and task-based parallelism

ECM3446/ECMM461: High Performance Computing

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Module Overview

Week 1: Introduction to HPC	Week 5: Floating point arithmetic	Week 8: Task parallelism and manager worker parallelism
Week 2: Introduction to OpenMP	Week 6: Memory and cache	Week 9: Linear algebra and matrices
Week 3: Numerical solutions to PDEs	Week 7: Factors affecting parallel performance	Week 10: Accelerators and GPUs
Week 4: Introduction to MPI		Week 11: Summary

Syllabus Plan

- Motivation of and introduction to high-performance computing
- Parallel computer architectures: shared-memory and distributed-memory architectures, multi-core processors, Graphics Processing Units (GPUs)
- Parallel programming methods: OpenMP, Message Passing Interface (MPI)
- Floating point arithmetic: floating point model, range, accuracy, exceptions
- Parallel processing algorithm and programming design: domain decomposition, halo exchange, manager-worker, task-based parallelism
- Parallel performance: speed-up, efficiency, parallel overheads and scaling
- Interconnection networks in high performance computers: topologies, latency and bandwidth

Manager-worker and task-based parallelism

- So far we have seen two ways to distribute work in parallel:
 - Using OpenMP to parallelise loops
 - Using domain decomposition with MPI
- These are data parallel approaches: the processes/threads carry out the same operations on different data
- These are tried and tested methods for parallelising work in HPC applications but sometimes another approach is required

Manager-worker and task-based parallelism

- Manager-worker and task-based parallelism are alternative ways of distributing work which allow for more flexibility
- Manager-worker: one process is a manager which hands out work to worker processes. A worker processes carries out a unit of work and request more work when the current work is completed.
- In task-based parallelism the workload is divided into tasks which are carried out by different processes/threads. This could involve different tasks being carried out on the same data.

Lecture Units

- Unit 8.1: More MPI features
- Unit 8.2: Manager-worker Mandelbrot
- Unit 8.3: More OpenMP work sharing constructs
- Unit 8.4: OpenMP tasks

Workshop 8

- In this workshop we will look at MPI versions of the Mandelbrot set calculation
- We test different MPI versions which distribute the work in different ways and see which one performs best
- The workshop uses information from units 8.1 and 8.2